



Emergence, Onset timing, and Predictors of Probable Insomnia in Previously Low-Symptom Pregnant Individuals across Pregnancy and Postpartum



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The authors have no conflict of interests to declare

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INTRODUCTION

Perinatal insomnia is associated with adverse mental health and birth outcomes. Little is known about who is most at risk.

This study examined the emergence and course of probable insomnia from late pregnancy to one year postpartum among women with low insomnia symptoms (< 8 Insomnia Severity Index [ISI]) in late pregnancy (26-32 weeks).

METHODS

Participants: 321 pregnant women (Age: 33.6 ± 3.8 years) with low insomnia symptoms at late pregnancy.

Outcome: Insomnia Symptoms (ISI) assessed at baseline, 5 weeks later, 3, 6 and 12 months postpartum.

Probable insomnia was categorised as ISI ≥ 10

Predictors: Age, Existing Children, Education, Dysfunctional Beliefs About Sleep, Sleep Reactivity (Ford Insomnia Response to Stress Test)

Analysis: Multinomial regressions, using the Never Insomnia group as the reference, examined predictors of trajectory.

RESULTS

Four trajectories were examined (**Figure 1**):

- Never Insomnia:** never met probable insomnia threshold,
- Insomnia Recovery:** met threshold in pregnancy and/or early postpartum but recovered by 12-months-postpartum;
- Persistent Insomnia:** met threshold in pregnancy and/or early postpartum which persisted at 12-months-postpartum;
- Late Postpartum-Onset Insomnia:** first met threshold at 12-months-postpartum.

Multinomial regressions shows, compared to “Never Insomnia” (**Figure 2**):

- Higher **DYSFUNCTIONAL BELIEFS ABOUT SLEEP** = **Insomnia Recovery**, **Persistent Insomnia** & **Late Postpartum-Onset**.
- Higher **SLEEP REACTIVITY** = **Persistent Insomnia** & **Insomnia Recovery**.
- Having existing children (**PARITY**) = **Persistent Insomnia**. **AGE** and **EDUCATION** were not associated with any trajectory.

CONCLUSION

Among individuals with low insomnia symptoms in late pregnancy, distinct insomnia trajectories emerge. **Dysfunctional sleep beliefs** and **sleep reactivity** may be key risk factors and having existing children may increase vulnerability to persistent symptoms. Targeting maladaptive sleep beliefs and identifying individuals with heightened sleep reactivity may offer important opportunities for early prevention of perinatal insomnia.

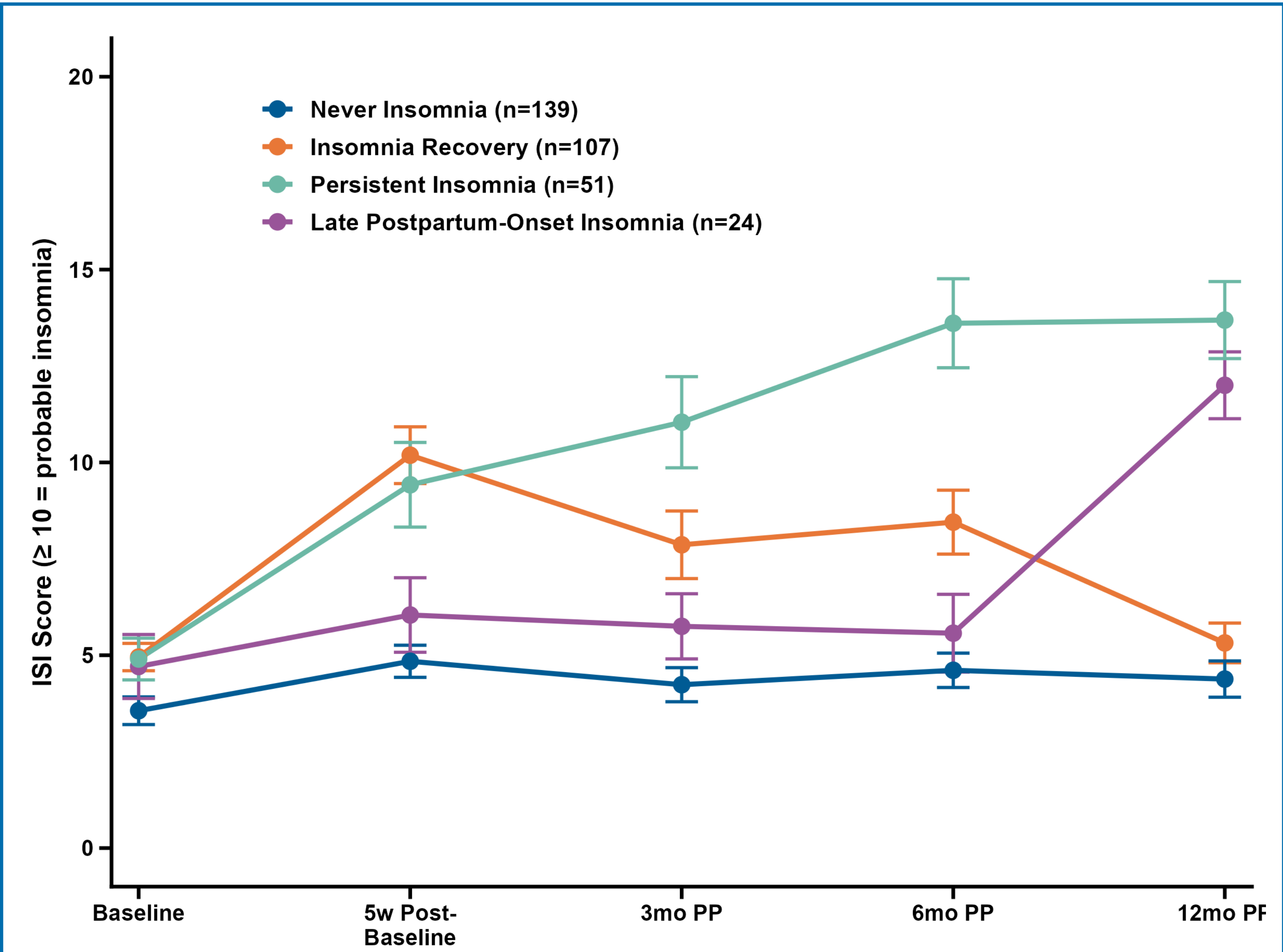


Figure 1. Trajectory of insomnia symptoms (ISI) from late pregnancy (26-32 weeks) to 12 months postpartum, colour coded by identified trajectory of probable insomnia.

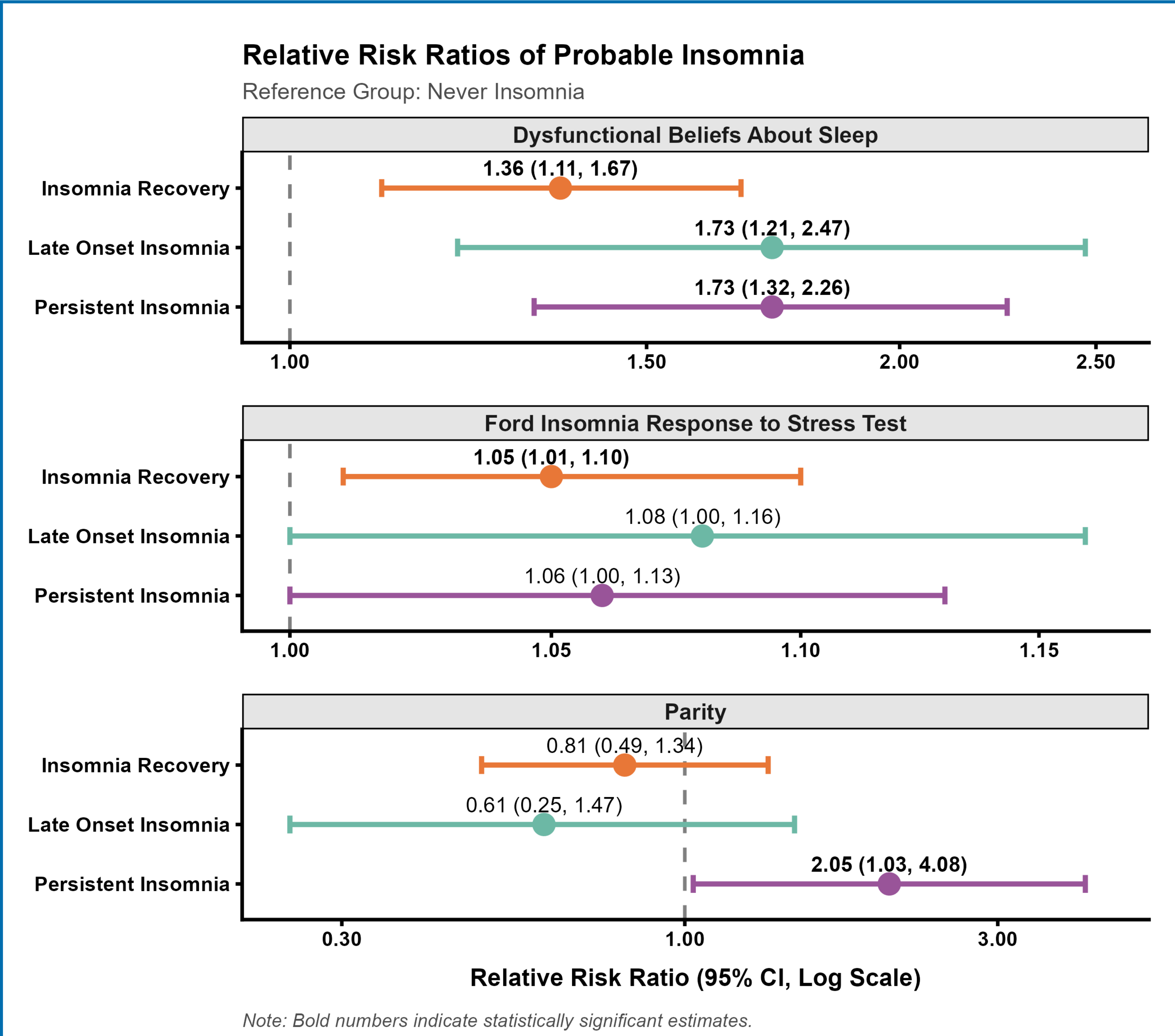


Figure 2. Forest plot of the Relative Risk Ratio & 95% Confidence Intervals for Multinomial Regressions of predictors of insomnia trajectory relative to “Never Insomnia”